
On the NP-Hardness of Feature-Weighted Gaussian c -Separation with Relaxed Selection

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Abstract

We investigate the complexity of Gaussian c -separation for sparse feature selection with diagonal class covariances. In the standard formulation, one selects exactly m features via a binary vector $\alpha \in \{0, 1\}^d$. A natural question is whether the continuous relaxation $\alpha \in [0, 1]^d$ yields a tractable problem. We answer this in the negative: the decision version of the relaxed problem remains NP-Hard. Our proof is a polynomial-time Karp reduction from EQUAL-CARDINALITY PARTITION restricted to perfect squares. The construction combines two components. First, a partition gadget encodes the balancing constraint through hinge-based separation terms. Second, a binary-well gadget forces every optimal relaxed solution to be integral under the cardinality constraint, despite the absence of explicit integrality restrictions. We then realize these weighted terms within the Gaussian c -separation model using class means and diagonal covariance vectors satisfying a monotone variance schedule. Beyond hardness, we identify two positive results. When the target slack threshold is zero, the relaxed problem reduces to a linear feasibility problem and is therefore solvable in polynomial time. In addition, for fixed dimension, the relaxed objective admits a polynomial-time approximation scheme. Taken together, these results show that continuous relaxation does not eliminate the intrinsic combinatorial hardness of Gaussian c -separation, while also delineating tractable boundary cases.

1 Introduction

Selecting a small set of marker genes that jointly distinguish all cell types is a fundamental task in single-cell data analysis [2, 5]. Several methods model this as a combinatorial optimization problem: scGeneFit [3] and G-PC [4] select genes so that pairwise distances between cell types exceed a threshold, while [7] uses set cover solved via integer linear programming. Even simplified formulations of this problem tend to be computationally hard. We focus on the Gaussian c -separation model of Borozan et al. [1], which captures pairwise class separation via Gaussian classes with diagonal covariances, and show that this problem is NP-complete.

A natural formulation uses a binary selector $\alpha \in \{0, 1\}^d$ of exactly m features. Following Borozan et al. [1], we study the *Gaussian c -separation* model with diagonal class covariances:

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given n Gaussian classes, minimize the total slack in achieving a prescribed separation margin $c > 0$ across an ordered set L of class pairs (see Eq. (1) for the formal statement).

The question. Borozan et al. [1] proved that the decision version is NP-complete in the binary case via a Turing reduction from PARTITION. A natural follow-up is whether relaxing $\alpha \in \{0, 1\}^d$ to the continuous hypercube $\alpha \in [0, 1]^d$ yields a tractable program - LP and QP relaxations frequently admit polynomial-time solutions or provably tight approximations in combinatorial optimization.

Our result. We show this is *not* the case. The problem remains NP-hard under the relaxation $\alpha \in [0, 1]^d$, even with diagonal Gaussian covariances. The reduction uses only $n = d + 4$ classes and $O(d)$ carefully scaled pairwise terms; no auxiliary decision variables or nonlinear constraints are introduced.

High-level idea. The reduction encodes EQUAL-CARDINALITY PARTITION in two layers.

Partition gadget: three class gadgets produce a two-sided hinge on the scalar statistic $\alpha^\top s$, so the objective reaches zero only when $\alpha^\top s = \frac{1}{2} \mathbf{1}^\top s$, directly encoding the partition equality.

Binary-well gadget: d per-coordinate gadgets contribute a strictly concave separable term $\sum_{k=1}^d -\frac{1}{1+\alpha_k}$ to the objective. Under the cardinality constraint $\alpha^\top \mathbf{1} = m$, this potential is minimized *uniquely* at the binary vertices $\alpha \in \{0, 1\}^d$, forcing integrality at every relaxed optimum without adding any integrality constraints.

Contributions.

- We prove that the *relaxed* Gaussian c -separation decision problem, with $\alpha \in [0, 1]^d$ and diagonal class covariances, is NP-hard.
- We give a polynomial-time Karp reduction from EQUAL-CARDINALITY PARTITION restricted to perfect squares. The reduction decomposes into a partition gadget, which enforces the balancing condition, and a binary-well gadget, which forces optimal relaxed solutions to be integral under the cardinality constraint.
- We provide an explicit realization of these abstract weighted gadgets within the Gaussian c -separation model using class means and diagonal covariance vectors that satisfy a monotone variance schedule.
- We also identify two tractable boundary cases: when the target threshold is $T = 0$, the relaxed problem reduces to a linear feasibility problem; and for fixed dimension, the relaxed objective admits a polynomial-time approximation scheme.

Paper outline. Section 2 defines Gaussian c -separation, recalls the decision formulation, and records basic properties of the model. Section 3 introduces the feature-weighted abstraction under a monotone variance schedule and develops the partition and binary-well gadgets. Section 4 combines these gadgets and proves NP-hardness of the relaxed decision problem. Section I presents the approximation result and the PTAS for fixed dimension. The appendices contain the supporting constructions and technical proofs, including NP-completeness of EQUAL-CARDINALITY PARTITION restricted to perfect squares, the monotone-schedule reduction, and the auxiliary gadget verifications.

2 Problem definition and properties

Fix the feature dimension $d \in \mathbb{N}$, the required sparsity $m \in \{1, \dots, d\}$, a positive margin $c > 0$, and a set L of ordered class pairs over n classes $1, \dots, n$. For each class i , let $\mu^{(i)} \in \mathbb{R}^d$ denote the mean and $\sigma^{(i)} \in \mathbb{R}_{>0}^d$ the vector of diagonal variances. The *Gaussian c -separation*

objective with slacks $\beta \in \mathbb{R}_{\geq 0}^L$ is

$$\begin{aligned}
& \min_{\alpha, \beta} \sum_{(i,j) \in L} \beta_{ij} \\
& \text{s.t.} \quad \sum_{k=1}^d \alpha_k (\mu_k^{(i)} - \mu_k^{(j)})^2 \geq (c - \beta_{ij}) \cdot \max\{\alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)}\}, \quad \forall (i,j) \in L, \\
& \quad \beta \geq 0, \quad \alpha^\top \mathbf{1} = m, \quad \alpha \in \{0, 1\}^d.
\end{aligned} \tag{1}$$

We study the associated *decision* problem: given a threshold $T \geq 0$, is there a feasible pair (α, β) with objective value at most T ?

NP-completeness proof. Borozan et al. [1] showed that problem (1) is NP-hard and its decision problem is NP-complete, by a reduction from PARTITION. In fact, we prove a stronger result by reducing from a restricted version of EQUAL-CARDINALITY PARTITION with all input numbers restricted to perfect squares (see Appendix A). This restricted variant is itself NP-complete, since standard PARTITION can be transformed into it in polynomial time, with only a polynomial increase in the encoding length (See Section A).

Consequently, by transitivity, we obtain a polynomial-time reduction from standard PARTITION whose output encoding length remains polynomially bounded in the input size, making it suitable for downstream analysis.

Theorem 1. *Let s be a vector of d perfect squares. Then there exist three d dimensional Gaussians that are $\frac{\|s\|}{2}$ -separated in dimension $\frac{d}{2}$, i.e., problem (1) admits a solution with objective value 0, if and only if s admits an equal-cardinality partition.*

Proof. Let s be a vector of d elements, each of which is a perfect square. If $s^\top \mathbf{1}$ is odd, then the instance clearly admits no partition. Hence, it suffices to consider the case in which $s^\top \mathbf{1}$ is even. Also if d is odd we cannot construct an equal-cardinality partition, hence we consider d is even.

Define

$$\begin{aligned}
\mu^{(1)} &:= \mathbf{0}, & \mu^{(2)} &:= m \sqrt{s} & \mu^{(3)} &:= -\frac{s^\top \mathbf{1}}{2} \mathbf{1}, \\
\sigma^{(1)} &:= \mathbf{1} & \sigma^{(2)} &:= m \mathbf{1} & \sigma^{(3)} &:= m s \\
c &:= \frac{s^\top \mathbf{1}}{2}, & m &:= \frac{d}{2}
\end{aligned}$$

Since the entries of s are perfect squares, the vector $\sqrt{s}$, elementwise square root of s , can be computed in polynomial time using only polynomially many additional bits. Moreover, $\frac{s^\top \mathbf{1}}{2}$ and $\frac{d}{2}$ are positive integers because $s^\top \mathbf{1}$ and d are even. All remaining quantities are obtained by addition and multiplication, and therefore also require only polynomially many additional bits. Consequently, this construction defines a polynomial-size reduction.

For this choice of Gaussians, the constraints become

$$\begin{aligned}
\sum_{k=1}^d \alpha_k (0 - m \sqrt{s_k})^2 &\geq \left(\frac{s^\top \mathbf{1}}{2} - \beta_{12} \right) \cdot \max\{\alpha^\top \mathbf{1}, m \alpha^\top \mathbf{1}\}, \\
\sum_{k=1}^d \alpha_k (0 - -\frac{s^\top \mathbf{1}}{2})^2 &\geq \left(\frac{s^\top \mathbf{1}}{2} - \beta_{13} \right) \cdot \max\{\alpha^\top \mathbf{1}, m \alpha^\top s\}, \\
\sum_{k=1}^d \alpha_k (m \sqrt{s_k} - -\frac{s^\top \mathbf{1}}{2})^2 &\geq \left(\frac{s^\top \mathbf{1}}{2} - \beta_{23} \right) \cdot \max\{m \alpha^\top \mathbf{1}, m \alpha^\top s\}.
\end{aligned}$$

Using $\mathbf{1}^\top \alpha = m$, these inequalities simplify to

$$\begin{aligned} m^2 s^\top \alpha &\geq \left(\frac{s^\top \mathbf{1}}{2} - \beta_{12} \right) m^2, \\ \left(\frac{s^\top \mathbf{1}}{2} \right)^2 m &\geq \left(\frac{s^\top \mathbf{1}}{2} - \beta_{13} \right) m \alpha^\top s, \\ m^2 s^\top \alpha + 2m \frac{s^\top \mathbf{1}}{2} (\sqrt{s} + \mathbf{1})^\top \alpha + \left(\frac{s^\top \mathbf{1}}{2} \right)^2 m &\geq \left(\frac{s^\top \mathbf{1}}{2} - \beta_{23} \right) m \alpha^\top s. \end{aligned}$$

Observe that the third inequality is implied by the second and $\beta_{23} \leq \beta_{13}$, hence the third inequality can be omitted.

We first prove that

$$\beta_{12} + \beta_{13} + \beta_{23} = 0 \implies \exists \text{ equal-cardinality partition of } s.$$

Since each $\beta_{ij} \geq 0$, the equality $\beta_{12} + \beta_{13} + \beta_{23} = 0$ implies that $\beta_{ij} = 0$ for all pairs (i, j) . Substituting $\beta_{23} = \beta_{13} = 0$ into the first two inequalities yields

$$\begin{aligned} s^\top \alpha &\geq \frac{s^\top \mathbf{1}}{2}, \\ \frac{s^\top \mathbf{1}}{2} &\geq \alpha^\top s. \end{aligned}$$

Hence,

$$s^\top \alpha = \frac{s^\top \mathbf{1}}{2},$$

which shows that α defines a partition of s with cardinality n .

For the converse direction, assume that s admits an equal-cardinality partition, i.e., $s^\top \alpha = n$. Then

$$s^\top \alpha = \frac{s^\top \mathbf{1}}{2}.$$

Substituting this identity into the first two inequalities gives

$$\begin{aligned} \frac{s^\top \mathbf{1}}{2} &\geq \frac{s^\top \mathbf{1}}{2} - \beta_{12}, \\ \left(\frac{s^\top \mathbf{1}}{2} \right)^2 &\geq \left(\frac{s^\top \mathbf{1}}{2} - \beta_{13} \right) \cdot \frac{s^\top \mathbf{1}}{2}. \end{aligned}$$

The minimal values of β_{12} and β_{23} satisfying these inequalities are therefore $\beta_{12} = \beta_{13} = 0$. Since $\beta_{23} \leq \beta_{13}$, it follows that $\beta_{23} = 0$ as well. Thus,

$$\beta_{12} + \beta_{13} + \beta_{23} = 0,$$

which completes the proof. $\square$

Corollary 1. *Decision version of Gaussian c -separation is NP-complete by Karp one-to-one reduction to equal-cardinality partition restricted to perfect squares.*

Proof. As shown in 1 we have a polynomial-time Karp reduction $f: (s) \mapsto (\mu^{(3)}, \sigma^{(3)}, \mu^{(2)}, \sigma^{(2)}, \mu^{(1)}, \sigma^{(1)}, c, m)$. Notice that given $\mu^{(2)}$ we can recover the original input s to the equal-cardinality partition problem by setting $s = (\frac{2}{d}\mu^{(2)})^{\circ 2}$, which implies $f(s) = f(s') \implies s = s'$, hence we have one-one reduction. $\square$

Relaxed c -separation special subproblem Checking if the gaussians are c -separable in the relaxed case is solvable in polynomial time, i.e., is there an $\alpha \in [0, 1]^d$ such that the value of the objective function is $T = 0$ is in P (See ??). Observe that by setting $\beta = 0$ we have that the relaxed version $\alpha \in [0, 1]^d$ becomes a feasibility problem of a linear program (LP), hence solvable in (weakly-)polynomial time. Unfortunately, the relaxed version is NP-complete under $T > 0$, which we will show in the following sections.

3 Feature-weighted abstraction

To simplify the exposition without loss of generality, we use a schedule with elementwise monotonically increasing variance vectors

$$\mathbf{0} < \sigma^{(1)} \leq \sigma^{(2)} \leq \dots \leq \sigma^{(n)} \quad (\text{elementwise}). \quad (2)$$

For any pair (i, j) with $i < j$, the right-hand side of Eq. (1) reduces to $(c - \beta_{ij}) \alpha^\top \sigma^{(i)}$. Eliminating the slacks yields the equivalent hinge loss

$$\min_{\alpha \in [0, 1]^d} \sum_{i=2}^n \sum_{j=1}^{i-1} \max \left\{ 0, c - \frac{\alpha^\top (\mu^{(i)} - \mu^{(j)})^{\circ 2}}{\alpha^\top \sigma^{(i)}} \right\} \quad \text{s.t. } \alpha^\top \mathbf{1} = m. \quad (3)$$

Proof deferred to Appendix B. Then partition problem can be rewritten in this form in the following way.

3.1 Encoding Partition: the two-hinge gadget

Let s denote a vector of d perfect squares in a EQUAL-CARDINALITY PARTITION restricted to perfect squares instance. Pick $c = \frac{s}{2}$. The choice of $\mu^{(i)}$ and $\sigma^{(i)}$ is invariant under translation and scaling hence we can modify them from Theorem 1 in the following way.

$$\begin{aligned} \mu^{(1)} &= \gamma(u + \mathbf{0}), & \mu^{(2)} &= \gamma(u + m\sqrt{s}), & \mu^{(3)} &= \gamma(u - \frac{s^\top \mathbf{1}}{2} \mathbf{1}), \\ \sigma^{(1)} &= \gamma^2 \mathbf{1}, & \sigma^{(2)} &= \gamma^2 m \mathbf{1}, & \sigma^{(3)} &= \gamma^2 m s, \end{aligned} \quad (4)$$

for any vector u and number $\gamma > 0$. With this construction we have

$$\begin{aligned} \frac{\alpha^\top (\mu^{(1)} - \mu^{(2)})^{\circ 2}}{\alpha^\top \sigma^{(2)}} &= \alpha^\top s, \\ \frac{\alpha^\top (\mu^{(1)} - \mu^{(3)})^{\circ 2}}{\alpha^\top \sigma^{(3)}} &= \frac{(s^\top \mathbf{1})^2}{4 \alpha^\top s}, \\ \frac{\alpha^\top (\mu^{(2)} - \mu^{(3)})^{\circ 2}}{\alpha^\top \sigma^{(3)}} &= \frac{(m\sqrt{s} + \frac{s^\top \mathbf{1}}{2} \mathbf{1})^{\circ 2}}{m s}. \end{aligned} \quad (5)$$

Then the contribution of these three classes equals

$$P(\alpha) := \max \left\{ 0, \frac{s^\top \mathbf{1}}{2} - \alpha^\top s \right\} + \max \left\{ 0, \frac{s^\top \mathbf{1}}{2} - \frac{(s^\top \mathbf{1})^2}{4 \alpha^\top s} \right\} + \max \left\{ 0, \frac{s^\top \mathbf{1}}{2} - \frac{\alpha^\top (m\sqrt{s} + \frac{s^\top \mathbf{1}}{2} \mathbf{1})^{\circ 2}}{m \alpha^\top s} \right\}. \quad (6)$$

By Theorem 1 we have that $P(\alpha) = 0 \iff \alpha^\top s = \frac{s^\top \mathbf{1}}{2}$, and $P(\alpha) > 0$ otherwise.

3.2 Forcing integrality at the optimum: the binary-well gadget

For each feature $k = 1, \dots, d$ we add a class $i = k$ whose contribution is $O\left(\frac{1}{1+\alpha_k}\right)$. This can be achieved by setting

$$\begin{aligned} \mu^{(0)} &:= \mathbf{1}, & \mu^{(i)} &:= m^i \mathbf{1}, & \forall i = 1, \dots, d, \\ \sigma^{(0)} &:= \mathbf{1}, & \sigma^{(i)} &:= \sum_{j=0}^{i-1} (m^i - m^j)^2 (m e_i + \mathbf{1}) & \forall i = 1, \dots, d, \end{aligned} \quad (7)$$

where e_i is the i -th canonical vector (See Appendix C for details). This gives us

$$\frac{\sum_{j < i} \alpha^\top (\mu^{(i)} - \mu^{(j)})^{\circ 2}}{\alpha^\top \sigma^{(i)}} = \frac{\alpha^\top \mathbf{1} \sum_{j=0}^{i-1} (m^i - m^j)^2}{(m \alpha^\top e_i + \alpha^\top \mathbf{1}) \sum_{j=1}^{i-1} (m^i - m^j)^2} = \frac{m}{m \alpha_i + m} = \frac{1}{1 + \alpha_i}. \quad (8)$$

Continuing, for each $\mu^{(i)}$ we have

$$\frac{\alpha^\top (\mu^{(i)} - \mu^{(j)})^{\circ 2}}{\alpha^\top \sigma^{(i)}} \leq \frac{\sum_{j=0}^{i-1} \alpha^\top (\mu^{(i)} - \mu^{(j)})^{\circ 2}}{\alpha^\top \sigma^{(i)}} = \frac{1}{1 + \alpha_i} \leq 1 \leq c,$$

hence

$$\begin{aligned}
B(\alpha) &:= \sum_{i=1}^d \sum_{j=0}^{i-1} \max \left\{ 0, c - \frac{\alpha^\top (\mu^{(i)} - \mu^{(j)})^{\circ 2}}{\alpha^\top \sigma^{(i)}} \right\} \\
&= \sum_{i=1}^d \sum_{j=0}^{i-1} c - \frac{\alpha^\top (\mu^{(i)} - \mu^{(j)})^{\circ 2}}{\alpha^\top \sigma^{(i)}} \\
&= \sum_{i=1}^d \left[ic - \frac{\sum_{j=0}^{i-1} \alpha^\top (\mu^{(i)} - \mu^{(j)})^{\circ 2}}{\alpha^\top \sigma^{(i)}} \right] \\
&= \sum_{i=1}^d ic - \frac{1}{1 + \alpha_i} \\
&= c \frac{d(d+1)}{2} - \sum_{i=1}^d \frac{1}{1 + \alpha_i}
\end{aligned} \tag{9}$$

The following lemma shows that this separable term is minimized exactly at the binary vertices.

Lemma 1 (Binary well). *Let $f(\alpha) := \sum_{k=1}^d -\frac{1}{1+\alpha_k}$ on the polytope $\{\alpha \in [0, 1]^d : \alpha^\top \mathbf{1} = m\}$ with integer $m \in \{0, \dots, d\}$. Then every minimizer of f is m -hot, i.e., $\alpha \in \{0, 1\}^d$ and $\alpha^\top \mathbf{1} = m$.*

Proof. Set $g(t) = -\frac{1}{1+t}$. Then $g'(t) = \frac{1}{(1+t)^2} > 0$ and $g''(t) = -\frac{2}{(1+t)^3} < 0$, so g is increasing and strictly concave on $[0, 1]$.

Fix any feasible $\alpha \in [0, 1]^d$ with $\sum_{i=1}^d \alpha_i = m$, and let $\alpha^\downarrow$ denote its decreasing rearrangement. Let $e \in \{0, 1\}^d$ be the m -hot vector sorted in decreasing order:

$$e = (\underbrace{1, \dots, 1}_{m \text{ times}}, \underbrace{0, \dots, 0}_{d-m \text{ times}}).$$

For every $k \in \{1, \dots, d\}$ we have

$$\sum_{i=1}^k \alpha_i^\downarrow \leq \min\{k, m\} = \sum_{i=1}^k e_i,$$

since each $\alpha_i \leq 1$ and $\sum_i \alpha_i = m$. Thus e majorizes α (i.e., $e \succ \alpha \succ \frac{m}{d} \mathbf{1}$).

By Karamata's inequality [6], for a concave function g and vectors with equal sum, majorization reverses the inequality:

$$\sum_{i=1}^d g(\alpha_i) \geq \sum_{i=1}^d g(e_i).$$

Equivalently, $f(\alpha) \geq f(e)$. Hence any minimizer of f must be a permutation of e , i.e., an m -hot vector. Since g is strictly concave, by Karamata's inequality we have $f(\alpha^\downarrow) = f(b^\downarrow) \implies \alpha^\downarrow = b^\downarrow$, hence the only minimizers of f are m -hot vectors. $\square$

Corollary 2. $\forall \alpha \in [0, 1]^d$

$$B(\alpha) \geq B(\alpha^*) = c \frac{d(d+1)}{2} - \left(m \frac{1}{2} + (d-m) \right)$$

where α^* is any m -hot vector. Equality holds if and only if α is m -hot.

Proof. Proof is immediate from Lemma 1. Note that there are exactly m ones in vector α^* which give m times $\frac{1}{2}$, and $d-m$ zeros which give $d-m$ times 1. $\square$

4 Putting the gadgets together

Choose $c = \frac{s^\top \mathbf{1}}{2}$. First we pick the binary well gadget from Eq. (7), i.e.,

$$\begin{aligned}\mu^{(i)} &:= \mathbf{1}, & \mu^{(i)} &:= m^i \mathbf{1}, & \forall i = 1, \dots, d, \\ \sigma^{(0)} &:= \mathbf{1}, & \sigma^{(i)} &:= \sum_{j=0}^{i-1} (m^i - m^j)^2 (m e_i + \mathbf{1}) & \forall i = 1, \dots, d.\end{aligned}$$

To make the second gadget not influence the first we will have to pick u and γ from Eq. (4) such that the gaussians are far enough and that the monotone variance schedule is met, e.g.,

$$\gamma := m + \sum_{j=0}^{d-1} (m^d - m^j)^2, \quad u := (s^\top \mathbf{1} + c + m^d).$$

Then, using $c = \frac{s^\top \mathbf{1}}{2}$ Eq. (4) becomes

$$\begin{aligned}\mu^{(d+1)} &= \gamma(3c + m^d) \mathbf{1}, & \mu^{(d+2)} &= \gamma((3c + m^d) \mathbf{1} + m \sqrt{s}), & \mu^{(d+3)} &= \gamma(2c + m^d) \mathbf{1}, \\ \sigma^{(d+1)} &= \gamma^2 \mathbf{1}, & \sigma^{(d+2)} &= \gamma^2 m \mathbf{1}, & \sigma^{(d+3)} &= \gamma^2 m s.\end{aligned} \quad (10)$$

By definition we have

$$\mathbf{0} < \sigma^{(0)} \leq \sigma^{(1)} \leq \dots \leq \sigma^{(d)} \leq \sigma^{(d+1)} \leq \sigma^{(d+2)} \leq \sigma^{(d+3)}.$$

Next, let us look at all $(i, j) \in \{d+1, d+2, d+3\} \times \{0, 1, \dots, d\}$. Notice that these gaussians are always well separated (See Appendix D), i.e.,

$$\max \left\{ 0, c - \frac{\alpha^\top (\mu^{(i)} - \mu^{(j)})^{\circ 2}}{\alpha^\top \sigma^{(i)}} \right\} = 0, \quad \forall (i, j) \in \{d+1, d+2, d+3\} \times \{0, 1, \dots, d\}.$$

With this choice of gaussians we get that Eq. (13) becomes

$$\min_{\alpha \in [0, 1]^d} P(\alpha) + B(\alpha), \quad \text{s.t. } \alpha^\top \mathbf{1} = m. \quad (11)$$

where $P(\alpha)$ is from Eq. (6) and $B(\alpha)$ is from Eq. (9).

Theorem 2. *The decision version of Gaussian c -separation under relaxed constraints is NP-hard via a polynomial-time Karp reduction from the EQUAL-CARDINALITY PARTITION problem restricted to perfect squares. Moreover, the reduction uses only polynomially many bits.*

Proof. Let $s \in \mathbb{N}_{>0}^d$ be in instance of EQUAL-CARDINALITY PARTITION restricted to perfect squares. Take $c = \frac{s}{2}$ and $m = \frac{d}{2}$. Build $\mu^{(i)}$ and $\sigma^{(i)}$ as shown above (Eqs. (7) and (10)). Since s are all perfect squares, $\sqrt{s}$ are all positive integers and can be calculated in polynomial time. $s^\top \mathbf{1}$ and d are even hence $\frac{s^\top \mathbf{1}}{2}$ and $\frac{d}{2}$ are integers. All other operations are multiplications and additions, of which we have polynomially many, hence we can build all $\mu^{(i)}$ and $\sigma^{(i)}$ in polynomial time. Then our Gaussian c -separation under relaxed constraints decision problem becomes "Does there exist α such that"

$$(P(\alpha) + B(\alpha) \leq T) \wedge \left(\alpha \in [0, 1]^d \right) \wedge \left(\alpha^\top \mathbf{1} = \frac{d}{2} \right)$$

Now, pick the minimum of Corollary 2

$$T = \frac{s^\top \mathbf{1}}{2} \frac{d(d+1)}{2} - \frac{3d}{4}$$

??? $\frac{3d}{4}$ can be rational. Is it worth it to multiply everything by 2 to get everything is an integer???. By Theorem 1 and Corollary 2 we have

$$P(\alpha) + B(\alpha) \geq 0 + \frac{s^\top \mathbf{1}}{2} \frac{d(d+1)}{2} - \frac{3d}{4}$$

and equality holds if and only if α is $\frac{d}{2}$ -hot and $s^\top \alpha = \frac{s^\top \mathbf{1}}{2}$. From this follows that $((\mu^{(i)}, \sigma^{(i)})_{i=0}^{d+4}, c, m, T)$ is a yes-instance of Gaussian c -separation with relaxed constraints if and only if s is a yes-instance of EQUAL-CARDINALITY PARTITION restricted to perfect squares, which proves Gaussian c -separation with relaxed constraints is NP-hard. To argue the bit complexity, let us look at the largest number, which is the element of either $\mu^{(d+2)}$ or $\sigma^{(d+3)}$. Denote $s_{max} := \max_{k=1, \dots, d} s_k$. Then

$$\begin{aligned} \max_{k=1, \dots, d} \mu_k^{(d+2)} &= \left(\frac{d}{2} + \sum_{i=0}^{d-1} \left(\left(\frac{d}{2} \right)^d - \left(\frac{d}{2} \right)^i \right)^2 \right) \left(s^\top \mathbf{1} + \frac{s^\top \mathbf{1}}{2} + \left(\frac{d}{2} \right)^d + \frac{d}{2} \sqrt{s_{max}} \right), \\ \max_{k=1, \dots, d} \sigma_k^{(d+3)} &= \left(\frac{d}{2} + \sum_{i=0}^{d-1} \left(\left(\frac{d}{2} \right)^d - \left(\frac{d}{2} \right)^i \right)^2 \right)^2 \frac{d}{2} s_{max}. \end{aligned}$$

The number of bits the largest number has is therefor

$$O(d \log d + \log s^\top \mathbf{1}).$$

We have $d(d+4) + 2$ many numbers. □

5 Polynomial-time Approximation algorithm

The idea is deriving a strict positive lower bound on the optimal value of the relaxed case and using the Lipschitz property approximating the solution. Given the problem

$$\begin{aligned} \min_{\alpha, \beta} \quad & \sum_{(i,j) \in L} \beta_{ij} \\ \text{s.t.} \quad & \sum_{k=1}^d \alpha_k (\mu_k^{(i)} - \mu_k^{(j)})^2 \geq (c - \beta_{ij}) \cdot \max\{\alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)}\}, \quad \forall (i,j) \in L, \\ & \beta \geq 0, \quad \alpha^\top \mathbf{1} = m, \quad \alpha \in [0, 1]^d. \end{aligned} \quad (12)$$

rewrite it into the hinge form

$$\begin{aligned} \min_{\alpha} \quad & \sum_{(i,j) \in L} \max \left\{ 0, c - \frac{\sum_{k=1}^d \alpha_k (\mu_k^{(i)} - \mu_k^{(j)})^2}{\max\{\alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)}\}} \right\} \\ \text{s.t.} \quad & \alpha^\top \mathbf{1} = m, \quad \alpha \in [0, 1]^d. \end{aligned} \quad (13)$$

For this problem to be well defined, and equivalent to Eq. (12) we have to have an assumption. Let

$$A := \{\alpha \in [0, 1]^d : \alpha^\top \mathbf{1} = m\}, \quad F_{ij}(\alpha) = \max \left\{ 0, c - \frac{\sum_{k=1}^d \alpha_k (\mu_k^{(i)} - \mu_k^{(j)})^2}{\max\{\alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)}\}} \right\},$$

$$\underline{\sigma} := \min_{\alpha \in A} \min_{(i,j) \in L} \max\{\alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)}\}.$$

With this notation we rewrite the problem to

$$\min_{\alpha \in A} \sum_{(i,j) \in L} F_{ij}(\alpha).$$

If $\underline{\sigma} > 0$ then all $F_{ij}(\alpha)$ are well defined, Eq. (13) also, and is equivalent to Eq. (12). Hence we further assume $\underline{\sigma} > 0$. Let us look at the separation gap,

$$\eta^* := \min_{\alpha \in A} \max_{(i,j) \in L} \max \left\{ 0, c \max\{\alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)}\} - \sum_{k=1}^d \alpha_k (\mu_k^{(i)} - \mu_k^{(j)})^2 \right\}. \quad (14)$$

Finding η^* is solvable in (weakly-)polynomial time by a LP solver. If there is no gap, i.e., $\eta^* = 0$, then the problem Eq. (12) can be solved with $\beta^* = 0$. Using this η^* we will derive a strict positive lower bound on the optimal value of the relaxed problem. If $\eta^* = 0$ we have an optimal α from the LP and we are finished. The tricky part is when $\eta^* > 0$.

Lemma 2. *Let η^* be the optimal value of Eq. (14) and α^* solution of the relaxed c -separated Gaussian problem. Define*

$$\delta := \frac{\eta^*}{\max_{i=1,\dots,n} \sum_{k=1}^d \sigma_k^{(i)}}.$$

Then

$$\sum_{(i,j) \in L} F_{ij}(\alpha^*) = 0 \iff \delta = 0$$

and

$$\sum_{(i,j) \in L} F_{ij}(\alpha^*) > 0 \iff \delta > 0.$$

We have our starting point. Next we show when the problem is Lipschitz continuous.

Proposition 1 (*G-Lipschitz property of the problem*). *If $\underline{\sigma} > 0$, then there exists a constat $G > 0$ such that the function $\sum_{(i,j) \in L} F_{ij}(\alpha)$ is G -Lipschitz on A .*

Using this we will partition the problem into equally distanced regions. For this we will need a polynomial bound on the fraction G/δ based on the bit length of the input.

Lemma 3 (*Bit-length of G/δ*). *Assume $\underline{\sigma} > 0$. Assume further that d is fixed and that all input data*

$$((\mu^{(i)}, \sigma^{(i)})_{i=1}^n, c, m, L)$$

are rational numbers encoded in binary. Let B denote the total bit-length of the instance, and let G and δ be the quantities from Proposition 1 and Lemma 2. If $\delta > 0$, then

$$\log_2 \left(\frac{G}{\delta} \right) = O_d(B^2).$$

Equivalently, the binary encoding length of G/δ is polynomially bounded in the input size.

Lemma 4 (*Explicit dyadic ℓ_1 -net of the hypersimplex*). *Let*

$$\mathcal{A} := \{\alpha \in [0, 1]^d : \alpha^\top \mathbf{1} = m\},$$

where $m \in \{0, \dots, d\}$ is an integer. For $q \in \mathbb{N}$, set $N := 2^q$ and define

$$\mathcal{N}_q := \left\{ \frac{1}{N} z : z \in \{0, 1, \dots, N\}^d, \sum_{i=1}^d z_i = Nm \right\}.$$

Then $\mathcal{N}_q \subseteq \mathcal{A}$, and for every $\alpha \in \mathcal{A}$ there exists $\bar{\alpha} \in \mathcal{N}_q$ such that

$$\|\alpha - \bar{\alpha}\|_1 \leq \frac{d}{2^q}.$$

In particular, $\mathcal{N}_q$ is an ℓ_1 -net of $\mathcal{A}$ with radius $d2^{-q}$.

Moreover, every point of $\mathcal{N}_q$ is rational with denominator 2^q , hence each coordinate has binary encoding length $O(q)$. The number of grid points satisfies

$$|\mathcal{N}_q| = [x^{m2^q}](1 + x + \dots + x^{2^q})^d \leq \binom{m2^q + d - 1}{d - 1}.$$

Consequently, if d is fixed and q is polynomially bounded in the input size, then the net can be represented and enumerated with polynomial bit complexity per point.

Theorem 3 (Grid-based approximation scheme in fixed dimension). *Assume $\sigma > 0$, and let*

$$F(\alpha) := \sum_{(i,j) \in L} F_{ij}(\alpha), \quad \mathcal{A} := \{\alpha \in [0, 1]^d : \alpha^\top \mathbf{1} = m\}.$$

Assume furthermore that d is fixed and that all input data are rational and encoded in binary. Let B denote the total bit-length of the instance.

Let G be the Lipschitz constant from Proposition 1, and let

$$\Sigma_{\max} := \max_{i=1, \dots, n} \sum_{k=1}^d \sigma_k^{(i)}.$$

Let η^ be the optimal value of Eq. (14). If $\eta^* = 0$, then $F^* = 0$ and an optimal solution can be obtained by solving the corresponding linear program. Suppose therefore that $\eta^* > 0$, and define*

$$\delta := \frac{\eta^*}{\Sigma_{\max}} > 0.$$

For $\varepsilon \in (0, 1)$, set

$$q := \left\lceil \log_2 \left(\frac{dG}{\varepsilon \delta} \right) \right\rceil, \quad N := 2^q,$$

and define the dyadic grid

$$\mathcal{N}_q := \left\{ \frac{z}{N} : z \in \{0, 1, \dots, N\}^d, \sum_{i=1}^d z_i = Nm \right\}.$$

Let

$$\hat{\alpha} \in \arg \min_{\alpha \in \mathcal{N}_q} F(\alpha).$$

Then

$$F(\hat{\alpha}) \leq (1 + \varepsilon)F^*, \quad F^* := \min_{\alpha \in \mathcal{A}} F(\alpha).$$

Moreover,

$$|\mathcal{N}_q| \leq \binom{m2^q + d - 1}{d - 1},$$

so the running time is

$$O(\text{poly}(B) + |L| d |\mathcal{N}_q|) = O_d \left(\text{poly}(B) + |L| \left(\frac{G}{\varepsilon \delta} \right)^{d-1} \right).$$

Corollary 3 (PTAS under a polynomial gap-ratio bound). *Under the assumptions of Theorem 3, by Lemma 3 there exists a polynomial p such that*

$$\frac{G}{\delta} \leq p(B)$$

for every instance in the family under consideration. Then the algorithm in Theorem 3 is a PTAS: for every $\varepsilon \in (0, 1)$, it returns a point $\hat{\alpha} \in \mathcal{A}$ satisfying

$$F(\hat{\alpha}) \leq (1 + \varepsilon)F^*,$$

and its running time is polynomial in B and $1/\varepsilon$.

Proof. By Theorem 3, the approximation guarantee already holds. It remains only to prove polynomial running time.

Since d is fixed and $G/\delta \leq p(B)$, we have

$$2^q \leq 2 \frac{dG}{\varepsilon \delta} \leq \frac{2d p(B)}{\varepsilon},$$

and therefore

$$|\mathcal{N}_q| \leq \binom{m2^q + d - 1}{d - 1} = O_d \left(\left(\frac{p(B)}{\varepsilon} \right)^{d-1} \right).$$

Because d is fixed, this is polynomial in B and $1/\varepsilon$. The full running time is therefore polynomial in B and $1/\varepsilon$, as required. $\square$

Approximation for the Binary Variant. Consider the original binary problem $A = \{\alpha \in \{0, 1\}^d : \alpha^\top \mathbf{1} = m\}$. Since the continuous feasible region contains the binary one, the continuous optimum satisfies $F_{\text{cont}}^* \leq F_{\text{bin}}^*$, so a multiplicative approximation for the continuous problem does not immediately imply the same guarantee for the binary problem. However, if $\hat{\alpha}$ is the $(1 + \varepsilon)$ -approximate continuous solution, we may round it to a feasible binary vector $R(\hat{\alpha})$ by setting the largest m coordinates of $\hat{\alpha}$ to 1 and the rest to 0. By G -Lipschitz continuity,

$$F(R(\hat{\alpha})) \leq F(\hat{\alpha}) + L\|R(\hat{\alpha}) - \hat{\alpha}\|_1.$$

Moreover, this top- m rounding satisfies

$$\|R(\hat{\alpha}) - \hat{\alpha}\|_1 \leq \frac{2m(d-m)}{d} \leq \frac{d}{2}.$$

Hence,

$$F(R(\hat{\alpha})) \leq (1 + \varepsilon)F_{\text{cont}}^* + G \frac{2m(d-m)}{d} \leq (1 + \varepsilon)F_{\text{bin}}^* + G \frac{2m(d-m)}{d}.$$

Therefore, the continuous approximation algorithm yields an additive approximation guarantee for the binary problem. If in addition $F_{\text{bin}}^* \geq \Delta_{\text{bin}} > 0$, then

$$F(R(\hat{\alpha})) \leq \left(1 + \varepsilon + \frac{2Gm(d-m)}{d\Delta_{\text{bin}}}\right) F_{\text{bin}}^*,$$

which gives a multiplicative guarantee under the extra lower-bound assumption.

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References

- [1] Bartol Borozan, Luka Borozan, Domagoj Ševerdija, Domagoj Matijević, and Stefan Canzar. Optimal marker genes for c-separated cell types. In Sriram Sankararaman, editor, *Research in Computational Molecular Biology*, pages 424–427, Cham, 2025. Springer Nature Switzerland.
- [2] Sijie Chen, Yanting Luo, Haoxiang Gao, Fanhong Li, Yixin Chen, Jiaqi Li, Renke You, Minsheng Hao, Haiyang Bian, Xi Xi, Wenrui Li, Weiyu Li, Mingli Ye, Qiuchen Meng, Ziheng Zou, Chen Li, Haochen Li, Yangyuan Zhang, Yanfei Cui, Lei Wei, Fufeng Chen, Xiaowo Wang, Hairong Lv, Kui Hua, Rui Jiang, and Xuegong Zhang. hECA: The cell-centric assembly of a cell atlas. *iScience*, 25(5):104318, 2022.
- [3] Bianca Dumitrascu, Soledad Villar, Dustin G. Mixon, and Barbara E. Engelhardt. Optimal marker gene selection for cell type discrimination in single cell analyses. *Nature Communications*, 12(1):1186, Feb 2021.
- [4] Euxhen Hasanaj, Amir Alavi, Anupam Gupta, Barnabás Póczos, and Ziv Bar-Joseph. Multiset multicover methods for discriminative marker selection. *Cell Reports Methods*, 2(11):100332, 2022.
- [5] Shuai He, Lin-He Wang, Yang Liu, Yi-Qi Li, Hai-Tian Chen, Jing-Hong Xu, Wan Peng, Guo-Wang Lin, Pan-Pan Wei, Bo Li, Xiaojun Xia, Dan Wang, Jin-Xin Bei, Xiaoshun He, and Zhiyong Guo. Single-cell transcriptome profiling of an adult human cell atlas of 15 major organs. *Genome Biology*, 21(1):294, Dec 2020.
- [6] Zoran Kadelburg, Dusan Dukic, Milivoje Lukic, and Ivan Matic. Inequalities of karamata, schur and muirhead, and some applications. *The Teaching of Mathematics*, 8(1):31–45, 2005.
- [7] Jonah Langlieb, Nina S. Sachdev, Karol S. Balderrama, Naeem M. Nadaf, Mukund Raj, Evan Murray, James T. Webber, Charles Vanderburg, Vahid Gazestani, Daniel Tward, Chris Mezas, Xu Li, Dylan M. Cable, Tabitha Norton, Partha Mitra, Fei Chen, and Evan Z. Macosko. The cell type composition of the adult mouse brain revealed by single cell and spatial genomics. *bioRxiv*, 2023.

A Equal-Cardinality Partition with perfect squares is NP-complete

$$\text{PARTITION} = \left\{ a_1, \dots, a_N \in \mathbb{N} \mid \exists I \subseteq [N], \sum_{i \in I} a_i = \sum_{i \notin I} a_i \right\}.$$

Equivalently,

$$(a_1, \dots, a_N) \in \text{PARTITION} \iff \exists \varepsilon_i \in \{\pm 1\} \text{ such that } \sum_{i=1}^N \varepsilon_i a_i = 0.$$

We reduce this problem to EQUAL-CARDINALITY PARTITION restricted to perfect squares, i.e.,

$$\text{EQUAL-CARDINALITY PARTITION} = \left\{ \sqrt{a_1}, \dots, \sqrt{a_N} \in \mathbb{N} \mid \exists I \subseteq [N], |I| = \frac{N}{2}, \sum_{i \in I} a_i = \sum_{i \notin I} a_i \right\}.$$

For $1 \leq i \leq N$, define

$$x_i^+ := (K^{N+i} + a_i K^{N-i})^2, \quad x_i^- := (K^{N+i} - a_i K^{N-i})^2,$$

where K is chosen large enough, e.g.

$$K := 2 \left(1 + \sum_{i=1}^N a_i^2 + 2 \sum_{i=1}^N a_i \right).$$

Each $x_i^\pm$ is a perfect square.

Expanding,

$$x_i^\pm = K^{2N+2i} \pm 2a_i K^{2N} + a_i^2 K^{2N-2i}.$$

Hence

$$x_i^+ + x_i^- = 2K^{2N+2i} + 2a_i^2 K^{2N-2i}.$$

Let

$$H := \frac{1}{2} \sum_{i=1}^N (x_i^+ + x_i^-) = \sum_{i=1}^N (K^{2N+2i} + a_i^2 K^{2N-2i}).$$

We claim that

$$(a_1, \dots, a_N) \in \text{PARTITION} \iff \exists S \subseteq \{x_1^+, x_1^-, \dots, x_N^+, x_N^-\} \text{ with } \sum_{x \in S} x = H \text{ and } |S| = N.$$

($\Rightarrow$) Assume $\sum_{i=1}^N \varepsilon_i a_i = 0$ for some $\varepsilon_i \in \{\pm 1\}$. Choose x_i^+ if $\varepsilon_i = +1$, and x_i^- if $\varepsilon_i = -1$. Then since exactly one of x_i^+ or x_i^- is chosen we have $|S| = N$, and

$$\sum_{i=1}^N (K^{2N+2i} + \varepsilon_i 2a_i K^{2N} + a_i^2 K^{2N-2i}) = H + 2K^{2N} \sum_{i=1}^N \varepsilon_i a_i = H.$$

So the square instance is a yes-instance.

($\Leftarrow$) Suppose a subset S sums to H . Write $c_i^\pm \in \{0, 1\}$ for whether $x_i^\pm \in S$. Then

$$\sum_{i=1}^N (c_i^+ x_i^+ + c_i^- x_i^-) = H,$$

so

$$0 = \sum_{i=1}^N (c_i^+ + c_i^- - 1) (K^{2N+2i} + a_i^2 K^{2N-2i}) + 2K^{2N} \sum_{i=1}^N (c_i^+ - c_i^-) a_i.$$

Set $m_i := c_i^+ + c_i^- - 1 \in \{-1, 0, 1\}$. If some $m_j \neq 0$, let j be maximal with this property. Then the term

$$m_j K^{2N+2j}$$

cannot be canceled by all smaller-order terms, because K was chosen sufficiently large. Contradiction. Thus $m_i = 0$ for all i , so for each i exactly one of x_i^+, x_i^- is chosen.

Therefore $c_i^+ - c_i^- \in \{\pm 1\}$, and the above equality reduces to

$$2K^{2N} \sum_{i=1}^N (c_i^+ - c_i^-) a_i = 0.$$

Hence

$$\sum_{i=1}^N \varepsilon_i a_i = 0 \quad \text{for} \quad \varepsilon_i := c_i^+ - c_i^- \in \{\pm 1\},$$

so the original PARTITION instance is also a yes-instance.

Therefore EQUAL-CARDINALITY PARTITION on perfect squares is NP-hard. It is in NP since a proposed subset can be verified in polynomial time. Hence it is NP-complete.

Size and running time. The number K has polynomially many bits in the input size. Since the largest exponent is $N + i \leq 2N$, each number $x_i^\pm$ has

$$O(N \log K)$$

bits, hence polynomial size. There are $2N$ such numbers, so the whole output is polynomially bounded. Computing powers, sums, differences, and squaring can all be done in polynomial time. Thus the reduction is a polynomial-time one-one Karp reduction.

The reduction is one-to-one.

Proposition 2. *The map $f: (a_1, \dots, a_N) \mapsto (\{x_1^+, x_1^-, \dots, x_N^+, x_N^-\}, H)$ is injective.*

Proof. We show the input can be recovered from the output. First, $N = |\text{output multiset}|/2$. Next, since $\sqrt{x_i^+} = K^{N+i} + a_i K^{N-i}$ and $\sqrt{x_{i+1}^-} = K^{N+i+1} - a_{i+1} K^{N-i-1}$, we have

$$\sqrt{x_{i+1}^-} - \sqrt{x_i^+} \geq K^{N+i}(K-1) - \left(\sum_j a_j\right)(K^{N-1} + 1) > 0$$

by the choice $K > 2\sum_j a_j$. Hence sorting the $2N$ values uniquely identifies each pair (x_i^-, x_i^+) in order. Finally, the identities

$$\sqrt{x_i^+} + \sqrt{x_i^-} = 2K^{N+i}, \quad a_i = \frac{\sqrt{x_i^+} - \sqrt{x_i^-}}{2K^{N-i}}$$

recover K (from $i = 1, 2$) and then each a_i uniquely. $\square$

B Monotone Schedule

Proposition 3. *Under the elementwise monotone variance schedule*

$$0 < \sigma^{(1)} \leq \sigma^{(2)} \leq \dots \leq \sigma^{(n)}, \quad (15)$$

the constraint in Eq. (1) simplifies, for every pair (i, j) with $i > j$, to

$$\alpha^\top (\mu^{(i)} - \mu^{(j)})^{\circ 2} \geq (c - \beta_{ij}) \alpha^\top \sigma^{(i)}.$$

Consequently, eliminating the slack variables β_{ij} yields the equivalent formulation

$$\min_{\alpha \in [0,1]^d} \sum_{i=1}^n \sum_{j=1}^{i-1} \max \left\{ 0, c - \frac{\alpha^\top (\mu^{(i)} - \mu^{(j)})^{\circ 2}}{\alpha^\top \sigma^{(i)}} \right\} \quad \text{s.t.} \quad \alpha^\top \mathbf{1} = m. \quad (16)$$

Proof. Fix (i, j) with $i > j$. By Eq. (15), the vector $\sigma^{(i)} - \sigma^{(j)}$ is elementwise non-negative. Since $\alpha \in [0, 1]^d$ is componentwise non-negative, it follows that

$$\alpha^\top (\sigma^{(i)} - \sigma^{(j)}) \geq 0,$$

and hence

$$\alpha^\top \sigma^{(i)} \geq \alpha^\top \sigma^{(j)}.$$

Therefore,

$$\max\{\alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)}\} = \alpha^\top \sigma^{(i)}.$$

Substituting this into Eq. (1) shows that, for every $i < j$, the corresponding constraint reduces to

$$\alpha^\top (\mu^{(i)} - \mu^{(j)})^{\circ 2} \geq (c - \beta_{ij}) \alpha^\top \sigma^{(i)}.$$

We now eliminate the slack variables. Fix any feasible α with $\alpha^\top \mathbf{1} = m$. Because $\sigma^{(i)} > \mathbf{0}$ elementwise and $\alpha \neq \mathbf{0}$, we have $\alpha^\top \sigma^{(i)} > 0$ for every i . Hence the preceding inequality is equivalent to

$$\beta_{ij} \geq c - \frac{\alpha^\top (\mu^{(i)} - \mu^{(j)})^{\circ 2}}{\alpha^\top \sigma^{(i)}}.$$

Combining this with the constraint $\beta_{ij} \geq 0$, we obtain

$$\beta_{ij} \geq \max\left\{0, c - \frac{\alpha^\top (\mu^{(i)} - \mu^{(j)})^{\circ 2}}{\alpha^\top \sigma^{(i)}}\right\}.$$

Since the objective in Eq. (1) is to minimize $\sum_{(i,j) \in L} \beta_{ij}$, the optimal choice of β_{ij} for fixed α is the smallest feasible value, namely

$$\beta_{ij}^* = \max\left\{0, c - \frac{\alpha^\top (\mu^{(i)} - \mu^{(j)})^{\circ 2}}{\alpha^\top \sigma^{(i)}}\right\}.$$

Substituting these optimal slacks back into the objective yields Eq. (16), proving the equivalence. $\square$

C Binary Well Justification

Proposition 4. *The choice of Eq. (7) satisfies the monotone variance schedule.*

Proof. Recall

$$\sigma^{(0)} := \mathbf{1}, \quad \sigma^{(i)} := \sum_{j=0}^{i-1} (m^i - m^j)^2 (\mathbf{1} + m e_i), \quad i = 1, \dots, d.$$

For $i \geq 1$, set

$$S_i := \sum_{j=0}^{i-1} (m^i - m^j)^2.$$

Then

$$\sigma^{(i)} = S_i (\mathbf{1} + m e_i),$$

so that each coordinate of $\sigma^{(i)}$ equals S_i , except the i -th coordinate, which equals $(m+1)S_i$.

We first derive a recursion for S_i . For $i \geq 2$,

$$\begin{aligned} S_i &= (m^i - 1)^2 + \sum_{j=1}^{i-1} (m^i - m^j)^2 \\ &= (m^i - 1)^2 + m^2 \sum_{j=0}^{i-2} (m^{i-1} - m^j)^2 \\ &= (m^i - 1)^2 + m^2 S_{i-1}. \end{aligned}$$

In particular, since $(m^i - 1)^2 \geq 0$, we obtain

$$S_i \geq m^2 S_{i-1}.$$

For $m \geq 2$, this implies $S_i \geq (m+1)S_{i-1}$.

We now compare $\sigma^{(i-1)}$ and $\sigma^{(i)}$ coordinatewise. Using the representation above,

$$\sigma^{(i-1)} = S_{i-1}(\mathbf{1} + m e_{i-1}), \quad \sigma^{(i)} = S_i(\mathbf{1} + m e_i).$$

Let $k \in \{1, \dots, d\}$. We distinguish three cases:

(i) $k \notin \{i-1, i\}$: Then

$$\sigma_k^{(i-1)} = S_{i-1} \leq S_i = \sigma_k^{(i)}.$$

(ii) $k = i-1$: Then

$$\sigma_{i-1}^{(i-1)} = (m+1)S_{i-1} \leq S_i = \sigma_{i-1}^{(i)},$$

where we used $S_i \geq (m+1)S_{i-1}$.

(iii) $k = i$: Then

$$\sigma_i^{(i-1)} = S_{i-1} \leq (m+1)S_i = \sigma_i^{(i)}.$$

Combining the three cases yields $\sigma^{(i-1)} \leq \sigma^{(i)}$ elementwise for all $i = 2, \dots, d$.

Finally, for $i = 1$,

$$\sigma^{(1)} = (m-1)^2(\mathbf{1} + m e_1) \geq \mathbf{1} = \sigma^{(0)}$$

for $m \geq 2$, which completes the proof. **?????? Here also ???????.** □

D Gadgets Do Not Influence Each Other

Proposition 5. Choose $c = \frac{s^\top \mathbf{1}}{2}$.

$$\mu^{(0)} := \mathbf{1}, \quad \mu^{(i)} := m^i \mathbf{1}, \quad \forall i = 1, \dots, d,$$

$$\sigma^{(0)} := \mathbf{1}, \quad \sigma^{(i)} := \sum_{j=0}^{i-1} (m^i - m^j)^2 (m e_i + \mathbf{1}) \quad \forall i = 1, \dots, d.$$

$$\gamma := m + \sum_{j=0}^{d-1} (m^d - m^j)^2, \quad u := (s^\top \mathbf{1} + c + m^d).$$

$$\begin{aligned} \mu^{(d+1)} &= \gamma(3c + m^d) \mathbf{1}, & \mu^{(d+2)} &= \gamma((3c + m^d) \mathbf{1} + m \sqrt{s}), & \mu^{(d+3)} &= \gamma(2c + m^d) \mathbf{1}, \\ \sigma^{(d+1)} &= \gamma^2 \mathbf{1}, & \sigma^{(d+2)} &= \gamma^2 m \mathbf{1}, & \sigma^{(d+3)} &= \gamma^2 m s. \end{aligned}$$

Then

$$\max \left\{ 0, c - \frac{\alpha^\top (\mu^{(i)} - \mu^{(j)})^{\circ 2}}{\alpha^\top \sigma^{(i)}} \right\} = 0, \quad \forall (i, j) \in \{d+1, d+2, d+3\} \times \{0, 1, \dots, d\}.$$

Proof. Going in order

$$\begin{aligned} c - \frac{\alpha^\top (\mu^{(d+1)} - \mu^{(i)})^{\circ 2}}{\alpha^\top \sigma^{(d+1)}} &= c - \frac{\alpha^\top (\gamma(3c + m^d) \mathbf{1} - m^i \mathbf{1})^{\circ 2}}{\gamma^2 \alpha^\top \mathbf{1}} \\ &\leq c - \frac{\alpha^\top (3\gamma c \mathbf{1})^{\circ 2}}{\gamma^2 \alpha^\top \mathbf{1}} \\ &= c - \frac{9c^2 \gamma^2 m}{\gamma^2 m} \leq 0, \end{aligned}$$

$$\begin{aligned}
c - \frac{\alpha^\top (\mu^{(d+2)} - \mu^{(i)})^{\circ 2}}{\alpha^\top \sigma^{(d+2)}} &= c - \frac{\alpha^\top (\gamma((3c + m^d)\mathbf{1} + m\sqrt{s}) - m^i\mathbf{1})^{\circ 2}}{\gamma^2 m \alpha^\top \mathbf{1}} \\
&\leq c - \frac{\alpha^\top (3\gamma c \mathbf{1} + \gamma m \sqrt{s})^{\circ 2}}{\gamma^2 m \alpha^\top \mathbf{1}} \\
&\leq c - \frac{\alpha^\top (3\gamma c \mathbf{1})^{\circ 2}}{\gamma^2 m \alpha^\top \mathbf{1}} \\
&= c - \frac{9\gamma^2 m c^2}{\gamma^2 m^2} \\
&= c - \frac{9c^2}{m} \\
&\leq c - \frac{9c^2}{2c} \leq 0, \\
\\
c - \frac{\alpha^\top (\mu^{(d+3)} - \mu^{(i)})^{\circ 2}}{\alpha^\top \sigma^{(d+3)}} &= c - \frac{\alpha^\top (\gamma(2c + m^d)\mathbf{1} - m^i\mathbf{1})^{\circ 2}}{\gamma^2 m \alpha^\top s} \\
&\leq c - \frac{\alpha^\top (2\gamma c \mathbf{1})^{\circ 2}}{\gamma^2 m \alpha^\top s} \\
&= c - \frac{4\gamma^2 m c^2}{\gamma^2 m \alpha^\top s} \\
&= c - \frac{4c^2}{\alpha^\top s} \\
&\leq c - \frac{4c^2}{2c} \leq 0,
\end{aligned}$$

hence the proof is concluded. $\square$

E Optimal Solution need not be rational

We show that even when all problem data are rational, the optimizer of (13) need not be a rational vector.

Proposition 6. *There exist rational vectors $\mu^{(i)}, \sigma^{(i)}$, a rational scalar c , and an integer m such that the optimization problem (13) admits a unique minimizer α^* with irrational entries.*

Proof. We construct an explicit example.

Let $d = 2$, $n = 3$, $m = 1$, and $c = 2$. Parametrize

$$\alpha = (x, 1 - x), \quad x \in [0, 1].$$

Choose

$$\mu^{(1)} = (0, 0), \quad \mu^{(2)} = (0, 1), \quad \mu^{(3)} = (1, 0),$$

so that

$$(\mu^{(2)} - \mu^{(1)})^{\circ 2} = (0, 1), \quad (\mu^{(3)} - \mu^{(1)})^{\circ 2} = (1, 0), \quad (\mu^{(3)} - \mu^{(2)})^{\circ 2} = (1, 1).$$

Let

$$\sigma^{(1)} = \sigma^{(2)} = (1, 2), \quad \sigma^{(3)} = (2, 2).$$

Then

$$\alpha^\top \sigma^{(1)} = \alpha^\top \sigma^{(2)} = 2 - x, \quad \alpha^\top \sigma^{(3)} = 2.$$

The three pairwise ratios in (13) become

$$\frac{\alpha^\top (\mu^{(2)} - \mu^{(1)})^{\circ 2}}{\max\{\alpha^\top \sigma^{(2)}, \alpha^\top \sigma^{(1)}\}} = \frac{1 - x}{2 - x},$$

$$\frac{\alpha^\top (\mu^{(3)} - \mu^{(1)})^{\circ 2}}{\max\{\alpha^\top \sigma^{(3)}, \alpha^\top \sigma^{(1)}\}} = \frac{x}{2},$$

$$\frac{\alpha^\top (\mu^{(3)} - \mu^{(2)})^{\circ 2}}{\max\{\alpha^\top \sigma^{(3)}, \alpha^\top \sigma^{(2)}\}} = \frac{1}{2}.$$

Since each of these quantities is strictly less than $c = 2$ for all $x \in [0, 1]$, all hinge terms are active. The objective therefore simplifies to

$$F(x) = \left(2 - \frac{1-x}{2-x}\right) + \left(2 - \frac{x}{2}\right) + \left(2 - \frac{1}{2}\right).$$

Minimizing $F(x)$ is equivalent to maximizing

$$g(x) = \frac{1-x}{2-x} + \frac{x}{2}.$$

We compute

$$g'(x) = -\frac{1}{(2-x)^2} + \frac{1}{2}.$$

Setting $g'(x) = 0$ gives

$$\frac{1}{2} = \frac{1}{(2-x)^2} \iff (2-x)^2 = 2,$$

so that

$$x = 2 - \sqrt{2},$$

since $2 + \sqrt{2} \notin [0, 1]$.

Moreover,

$$g''(x) = -\frac{2}{(2-x)^3} < 0 \quad \text{for all } x \in [0, 1],$$

so g is strictly concave and this critical point is the unique maximizer. Hence the unique minimizer of (13) is

$$\alpha^* = (2 - \sqrt{2}, \sqrt{2} - 1),$$

which is irrational. □

F Approximation Algorithm

Lemma 5. *Let η^* be the optimal value of Eq. (14) and α^* solution of the relaxed c -separated Gaussian problem. Define*

$$\delta := \frac{\eta^*}{\max_{i=1, \dots, n} \sum_{k=1}^d \sigma_k^{(i)}}.$$

Then

$$\sum_{(i,j) \in L} F_{ij}(\alpha^*) = 0 \iff \delta = 0$$

and

$$\sum_{(i,j) \in L} F_{ij}(\alpha^*) > 0 \iff \delta > 0.$$

Proof. For $\alpha \in A$ and $(i, j) \in L$ we have

$$F_{ij}(\alpha) \cdot \max\{\alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)}\} = c \cdot \max\{\alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)}\} - \sum_{k=1}^d \alpha_k (\mu_k^{(i)} - \mu_k^{(j)})^2.$$

and

$$\begin{aligned}
\left(\max_{(i,j) \in L} F_{ij}(\alpha) \right) \left(\max_{(i,j) \in L} \max \left\{ \alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)} \right\} \right) &\geq \max_{(i,j) \in L} F_{ij}(\alpha) \cdot \max \left\{ \alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)} \right\} \\
&= \max_{(i,j) \in L} c \cdot \max \left\{ \alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)} \right\} - \sum_{k=1}^d \alpha_k (\mu_k^{(i)} - \mu_k^{(j)})^2 \\
&\geq \eta^*,
\end{aligned}$$

hence

$$\sum_{(i,j) \in L} F_{ij}(\alpha) \geq \max_{(i,j) \in L} F_{ij}(\alpha) \geq \frac{\eta^*}{\max_{(i,j) \in L} \max \left\{ \alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)} \right\}} \geq \delta.$$

To prove implications, look at the definition of δ . $\delta = 0$ if and only if η^* and $\eta^* = 0$ if and only if $c = \frac{\sum_{k=1, \dots, d} \alpha_k (\mu_k^{(i)} - \mu_k^{(j)})^2}{\max \left\{ \alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)} \right\}}, \forall (i, j) \text{ in } L$, which holds if and only if $F_{ij}(\alpha^*) = 0, \forall (i, j) \in L$. $\square$

Proposition 7 (*G-Lipschitz property of the problem*). *If $\underline{\sigma} > 0$, then there exists a constat $G > 0$ such that the function $\sum_{(i,j) \in L} F_{ij}(\alpha)$ is G -Lipschitz on A .*

Proof. Let $\underline{\sigma} > 0$ and

$$\begin{aligned}
q_{ij}(a) &:= a^\top (\mu^{(i)} - \mu^{(j)})^{\circ 2}, \quad s_{ij}(a) := \max \left\{ a^\top \sigma^{(i)}, a^\top \sigma^{(j)} \right\}, \\
Q_{ij} &:= \max_{k=1, \dots, k} (\mu_k^{(i)} - \mu_k^{(j)})^2, \quad S_{ij} := \max_{k=1, \dots, d} \max \left\{ \sigma_k^{(i)}, \sigma_k^{(j)} \right\}.
\end{aligned}$$

For any $a, b \in A$, using the fact that $x \mapsto \max \{0, x\}$ is 1-Lipschitz,

$$\begin{aligned}
|F_{ij}(a) - F_{ij}(b)| &\leq \left| \frac{q_{ij}(a)}{s_{ij}(a)} - \frac{q_{ij}(b)}{s_{ij}(b)} \right| \\
&\leq \frac{|q_{ij}(a) - q_{ij}(b)|}{\underline{\sigma}} + \frac{|q_{ij}(b)| |s_{ij}(a) - s_{ij}(b)|}{(\underline{\sigma})^2}.
\end{aligned}$$

Moreover,

$$|q_{ij}(a) - q_{ij}(b)| \leq Q_{ij} \|a - b\|_1, \quad |q_{ij}(b)| \leq m Q_{ij}, \quad |s_{ij}(a) - s_{ij}(b)| \leq S_{ij} \|a - b\|_1.$$

Therefore

$$|F_{ij}(a) - F_{ij}(b)| \leq \left(\frac{Q_{ij}}{\underline{\sigma}} + \frac{m Q_{ij} S_{ij}}{\underline{\sigma}^2} \right) \|a - b\|_1.$$

Summing over $(i, j) \in L$ yields

$$\left| \sum_{(i,j) \in L} F_{ij}(a) - F_{ij}(b) \right| \leq \left[\sum_{(i,j) \in L} \left(\frac{Q_{ij}}{\underline{\sigma}} + \frac{m Q_{ij} S_{ij}}{\underline{\sigma}^2} \right) \right] \|a - b\|_1.$$

Thus $\sum_{(i,j) \in L} F_{ij}(\alpha)$ is G -Lipschitz on A w.r.t. $\|\cdot\|_1$, with

$$G = \sum_{(i,j) \in L} \left(\frac{Q_{ij}}{\underline{\sigma}} + \frac{m Q_{ij} S_{ij}}{\underline{\sigma}^2} \right).$$

$\square$

Proposition 8 (*G-Lipschitz property of the problem*). *If $\underline{\sigma} > 0$, then there exists a constat $G > 0$ such that the function $\sum_{(i,j) \in L} F_{ij}(\alpha)$ is G -Lipschitz on A .*

Proof. Let $\underline{\sigma} > 0$ and

$$q_{ij}(a) := a^\top (\mu^{(i)} - \mu^{(j)})^{\circ 2}, \quad s_{ij}(a) := \max \left\{ a^\top \sigma^{(i)}, a^\top \sigma^{(j)} \right\},$$

$$Q_{ij} = \max_{k=1,\dots,d} (\mu_k^{(i)} - \mu_k^{(j)})^2, \quad S_{ij} := \max_{k=1,\dots,d} \max \left\{ \sigma_k^{(i)}, \sigma_k^{(j)} \right\}.$$

For any $a, b \in A$, using the fact that $x \mapsto \max \{0, x\}$ is 1-Lipschitz,

$$\begin{aligned} |F_{ij}(a) - F_{ij}(b)| &\leq \left| \frac{q_{ij}(a)}{s_{ij}(a)} - \frac{q_{ij}(b)}{s_{ij}(b)} \right| \\ &\leq \frac{|q_{ij}(a) - q_{ij}(b)|}{\underline{\sigma}} + \frac{|q_{ij}(b)| |s_{ij}(a) - s_{ij}(b)|}{(\underline{\sigma})^2}. \end{aligned}$$

Moreover,

$$|q_{ij}(a) - q_{ij}(b)| \leq Q_{ij} \|a - b\|_1, \quad |q_{ij}(b)| \leq m Q_{ij}, \quad |s_{ij}(a) - s_{ij}(b)| \leq S_{ij} \|a - b\|_1.$$

Therefore

$$|F_{ij}(a) - F_{ij}(b)| \leq \left(\frac{Q_{ij}}{\underline{\sigma}} + \frac{m Q_{ij} S_{ij}}{\underline{\sigma}^2} \right) \|a - b\|_1.$$

Summing over $(i, j) \in L$ yields

$$\left| \sum_{(i,j) \in L} F_{ij}(a) - F_{ij}(b) \right| \leq \left[\sum_{(i,j) \in L} \left(\frac{Q_{ij}}{\underline{\sigma}} + \frac{m Q_{ij} S_{ij}}{\underline{\sigma}^2} \right) \right] \|a - b\|_1.$$

Thus $\sum_{(i,j) \in L} F_{ij}(\alpha)$ is G -Lipschitz on A w.r.t. $\|\cdot\|_1$, with

$$G = \sum_{(i,j) \in L} \left(\frac{Q_{ij}}{\underline{\sigma}} + \frac{m Q_{ij} S_{ij}}{\underline{\sigma}^2} \right).$$

□

Lemma 6 (Bit-length of G/δ). *Assume $\underline{\sigma} > 0$. Assume further that d is fixed and that all input data*

$$((\mu^{(i)}, \sigma^{(i)})_{i=1}^n, c, m, L)$$

are rational numbers encoded in binary. Let B denote the total bit-length of the instance, and let G and δ be the quantities from Proposition 1 and Lemma 2. If $\delta > 0$, then

$$\log_2 \left(\frac{G}{\delta} \right) = O_d(B^2).$$

Equivalently, the binary encoding length of G/δ is polynomially bounded in the input size.

Proof. For $(i, j) \in L$, define

$$Q_{ij} := \max_{k=1,\dots,d} (\mu_k^{(i)} - \mu_k^{(j)})^2, \quad S_{ij} := \max_{k=1,\dots,d} \max \{ \sigma_k^{(i)}, \sigma_k^{(j)} \}.$$

Since the input is rational and has total bit-length B , all numbers Q_{ij} , S_{ij} , and

$$\Sigma_{\max} := \max_{i=1,\dots,n} \sum_{k=1}^d \sigma_k^{(i)}$$

have binary length $O(B)$. Also $|L| \leq B$.

First we bound $\underline{\sigma}$ from below. For each fixed pair $(i, j) \in L$, let

$$\underline{\sigma}_{ij} := \min_{\alpha \in \mathcal{A}} \max \{ \alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)} \}.$$

This is the optimum of the rational linear program

$$\min_{\alpha, t} t \quad \text{s.t.} \quad t \geq \alpha^\top \sigma^{(i)}, \quad t \geq \alpha^\top \sigma^{(j)}, \quad \alpha \in \mathcal{A}.$$

Since d is fixed, this LP has $O_d(1)$ variables and constraints, and total encoding length $O_d(B)$. By the standard size bound for rational linear programs, every positive optimum has binary length $O_d(B^2)$. Hence, whenever $\underline{\sigma}_{ij} > 0$,

$$\log_2 \left(\frac{1}{\underline{\sigma}_{ij}} \right) = O_d(B^2).$$

Because

$$\underline{\sigma} = \min_{(i,j) \in L} \underline{\sigma}_{ij} > 0,$$

the same bound holds for $\underline{\sigma}$:

$$\log_2 \left(\frac{1}{\underline{\sigma}} \right) = O_d(B^2).$$

By Proposition 1,

$$G = \sum_{(i,j) \in L} \left(\frac{Q_{ij}}{\underline{\sigma}} + \frac{m Q_{ij} S_{ij}}{\underline{\sigma}^2} \right),$$

and therefore

$$\log_2 G = O_d(B^2).$$

Next, by definition,

$$\delta = \frac{\eta^*}{\Sigma_{\max}},$$

where η^* is the optimum value of the linear program in Eq. (14). Again, since d is fixed, this LP has total encoding length $O_d(B)$. If $\delta > 0$, then $\eta^* > 0$, and the same LP size bound gives

$$\log_2 \left(\frac{1}{\eta^*} \right) = O_d(B^2).$$

Since $\log_2 \Sigma_{\max} = O(B)$, it follows that

$$\log_2 \left(\frac{1}{\delta} \right) = \log_2 \Sigma_{\max} + \log_2 \left(\frac{1}{\eta^*} \right) = O_d(B^2).$$

Combining the bounds for G and δ , we obtain

$$\log_2 \left(\frac{G}{\delta} \right) = \log_2 G + \log_2 \left(\frac{1}{\delta} \right) = O_d(B^2),$$

as claimed. □

Lemma 7 (Explicit dyadic ℓ_1 -net of the hypersimplex). *Let*

$$\mathcal{A} := \{\alpha \in [0, 1]^d : \alpha^\top \mathbf{1} = m\},$$

where $m \in \{0, \dots, d\}$ is an integer. For $q \in \mathbb{N}$, set $N := 2^q$ and define

$$\mathcal{N}_q := \left\{ \frac{z}{N} : z \in \{0, 1, \dots, N\}^d, \sum_{i=1}^d z_i = Nm \right\}.$$

Then $\mathcal{N}_q \subseteq \mathcal{A}$, and for every $\alpha \in \mathcal{A}$ there exists $\bar{\alpha} \in \mathcal{N}_q$ such that

$$\|\alpha - \bar{\alpha}\|_1 \leq \frac{d}{2^q}.$$

In particular, $\mathcal{N}_q$ is an ℓ_1 -net of $\mathcal{A}$ with radius $d 2^{-q}$.

Moreover, every point of $\mathcal{N}_q$ is rational with denominator 2^q , hence each coordinate has binary encoding length $O(q)$. The number of grid points satisfies

$$|\mathcal{N}_q| = [x^{m2^q}](1 + x + \dots + x^{2^q})^d \leq \binom{m2^q + d - 1}{d - 1}.$$

Consequently, if d is fixed and q is polynomially bounded in the input size, then the net can be represented and enumerated with polynomial bit complexity per point.

Proof. Fix $\alpha \in \mathcal{A}$ and let $t_i := 2^q \alpha_i$. Then

$$t_i \in [0, 2^q], \quad \sum_{i=1}^d t_i = m2^q.$$

Write $r_i := \lfloor t_i \rfloor$, and set

$$R := m2^q - \sum_{i=1}^d r_i.$$

Since $m2^q$ is an integer, $R \in \{0, \dots, d\}$. Choose the R coordinates with largest fractional parts $t_i - r_i$, add 1 to those r_i 's, and call the resulting vector z . Then $z_i \in \{0, \dots, 2^q\}$ for all i , and

$$\sum_{i=1}^d z_i = m2^q.$$

Hence $\bar{\alpha} := z/2^q$ belongs to $\mathcal{N}_q$. Also, for every i ,

$$|z_i - t_i| \leq 1,$$

so

$$\|\bar{\alpha} - \alpha\|_1 = \frac{1}{2^q} \sum_{i=1}^d |z_i - t_i| \leq \frac{d}{2^q}.$$

This proves the net property.

Each coordinate of $\bar{\alpha} \in \mathcal{N}_q$ is of the form $z_i/2^q$, so it is a dyadic rational with binary encoding length $O(q)$.

Finally, $|\mathcal{N}_q|$ is exactly the number of integer solutions of

$$z_1 + \dots + z_d = m2^q, \quad 0 \leq z_i \leq 2^q,$$

which is the coefficient of x^{m2^q} in $(1 + x + \dots + x^{2^q})^d$. Dropping the upper bounds $z_i \leq 2^q$ yields the estimate

$$|\mathcal{N}_q| \leq \binom{m2^q + d - 1}{d - 1}.$$

□

Theorem 4 (Grid-based approximation scheme in fixed dimension). *Assume $\underline{\sigma} > 0$, and let*

$$F(\alpha) := \sum_{(i,j) \in L} F_{ij}(\alpha), \quad \mathcal{A} := \{\alpha \in [0, 1]^d : \alpha^\top \mathbf{1} = m\}.$$

Assume furthermore that d is fixed and that all input data are rational and encoded in binary. Let B denote the total bit-length of the instance.

Let G be the Lipschitz constant from Proposition 1, and let

$$\Sigma_{\max} := \max_{i=1, \dots, n} \sum_{k=1}^d \sigma_k^{(i)}.$$

Let η^ be the optimal value of Eq. (14). If $\eta^* = 0$, then $F^* = 0$ and an optimal solution can be obtained by solving the corresponding linear program. Suppose therefore that $\eta^* > 0$, and define*

$$\delta := \frac{\eta^*}{\Sigma_{\max}} > 0.$$

For $\varepsilon \in (0, 1)$, set

$$q := \left\lceil \log_2 \left(\frac{dG}{\varepsilon \delta} \right) \right\rceil, \quad N := 2^q,$$

and define the dyadic grid

$$\mathcal{N}_q := \left\{ \frac{z}{N} : z \in \{0, 1, \dots, N\}^d, \sum_{i=1}^d z_i = Nm \right\}.$$

Let

$$\hat{\alpha} \in \arg \min_{\alpha \in \mathcal{N}_q} F(\alpha).$$

Then

$$F(\hat{\alpha}) \leq (1 + \varepsilon)F^*, \quad F^* := \min_{\alpha \in \mathcal{A}} F(\alpha).$$

Moreover,

$$|\mathcal{N}_q| \leq \binom{m2^q + d - 1}{d - 1},$$

so the running time is

$$O(\text{poly}(B) + |L| d |\mathcal{N}_q|) = O_d \left(\text{poly}(B) + |L| \left(\frac{G}{\varepsilon \delta} \right)^{d-1} \right).$$

Proof. If $\eta^* = 0$, then by ?? the optimum value is $F^* = 0$, and an optimal solution is obtained by solving the linear program defining η^* . So assume $\eta^* > 0$. Then $\delta > 0$ by definition.

Let $\alpha^* \in \mathcal{A}$ be an optimizer of F . By Lemma 4, there exists $\bar{\alpha} \in \mathcal{N}_q$ such that

$$\|\bar{\alpha} - \alpha^*\|_1 \leq \frac{d}{2^q}.$$

By the choice of q ,

$$\frac{d}{2^q} \leq \frac{\varepsilon \delta}{G}.$$

Therefore, by Proposition 1,

$$F(\bar{\alpha}) \leq F(\alpha^*) + G \|\bar{\alpha} - \alpha^*\|_1 \leq F^* + \varepsilon \delta.$$

Since $\hat{\alpha}$ minimizes F over $\mathcal{N}_q$,

$$F(\hat{\alpha}) \leq F(\bar{\alpha}) \leq F^* + \varepsilon \delta.$$

By Lemma 2, every feasible point satisfies $F(\alpha) \geq \delta$, hence in particular $F^* \geq \delta$. Thus

$$F(\hat{\alpha}) \leq F^* + \varepsilon \delta \leq (1 + \varepsilon)F^*.$$

For the running time, Lemma 4 gives

$$|\mathcal{N}_q| \leq \binom{m2^q + d - 1}{d - 1}.$$

Evaluating $F(\alpha)$ at one grid point takes $O(|L|d)$ arithmetic operations, so exhaustive search over the grid costs

$$O(|L|d |\mathcal{N}_q|).$$

The preprocessing needed to compute η^* , δ , and G is polynomial in B for fixed d , giving the claimed bound. $\square$

G Linearized Approximation

Theorem 5. *Let*

$$F(\alpha) := \sum_{(i,j) \in L} \left[c - \frac{q_{ij}(\alpha)}{s_{ij}(\alpha)} \right]_+, \quad q_{ij}(\alpha) := \sum_{k=1}^d \alpha_k (\mu_k^{(i)} - \mu_k^{(j)})^2, \quad s_{ij}(\alpha) := \max\{\alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)}\},$$

on

$$A := \{\alpha \in [0, 1]^d : \alpha^\top \mathbf{1} = m\},$$

and assume

$$\underline{s} := \min_{\alpha \in A} \min_{(i,j) \in L} s_{ij}(\alpha) > 0, \quad \bar{s} := \max_{\alpha \in A} \max_{(i,j) \in L} s_{ij}(\alpha) < \infty.$$

Further, let

$$\eta^* := \min_{\alpha \in A} \max_{(i,j) \in L} [c s_{ij}(\alpha) - q_{ij}(\alpha)]_+,$$

and let α_η be an optimizer. Then, with

$$F^* := \min_{\alpha \in A} F(\alpha),$$

the point α_η satisfies

$$F(\alpha_\eta) \leq \frac{|L| \eta^*}{\underline{s}}, \quad F^* \geq \frac{\eta^*}{\bar{s}}.$$

Consequently,

$$F(\alpha_\eta) \leq |L| \frac{\bar{s}}{\underline{s}} F^*.$$

If, in addition, F is G -Lipschitz on A with respect to $\|\cdot\|_1$, then for every minimizer $\alpha^* \in \arg \min_A F$,

$$F(\alpha_\eta) \leq F^* + G \|\alpha_\eta - \alpha^*\|_1.$$

Hence, if $F^* \geq \delta > 0$ and $\|\alpha_\eta - \alpha^*\|_1 \leq \varepsilon \delta / G$, then

$$F(\alpha_\eta) \leq (1 + \varepsilon) F^*.$$

Proof. For every $(i, j) \in L$,

$$[c s_{ij}(\alpha_\eta) - q_{ij}(\alpha_\eta)]_+ \leq \eta^*,$$

so

$$F(\alpha_\eta) = \sum_{(i,j) \in L} \frac{[c s_{ij}(\alpha_\eta) - q_{ij}(\alpha_\eta)]_+}{s_{ij}(\alpha_\eta)} \leq \sum_{(i,j) \in L} \frac{\eta^*}{\underline{s}} = \frac{|L| \eta^*}{\underline{s}}.$$

Also, for any $\alpha \in A$,

$$F(\alpha) = \sum_{(i,j) \in L} \frac{[c s_{ij}(\alpha) - q_{ij}(\alpha)]_+}{s_{ij}(\alpha)} \geq \frac{\max_{(i,j) \in L} [c s_{ij}(\alpha) - q_{ij}(\alpha)]_+}{\bar{s}}.$$

Taking the minimum over $\alpha \in A$ gives

$$F^* \geq \frac{\eta^*}{\bar{s}}.$$

Combining the two bounds yields

$$F(\alpha_\eta) \leq |L| \frac{\bar{s}}{\underline{s}} F^*.$$

If F is G -Lipschitz, then

$$F(\alpha_\eta) \leq F(\alpha^*) + G \|\alpha_\eta - \alpha^*\|_1 = F^* + G \|\alpha_\eta - \alpha^*\|_1.$$

The final claim follows from $F^* \geq \delta$ and $\|\alpha_\eta - \alpha^*\|_1 \leq \varepsilon \delta / G$. $\square$

Theorem 6. *Let*

$$F(\alpha) := \sum_{(i,j) \in L} \left[c - \frac{q_{ij}(\alpha)}{s_{ij}(\alpha)} \right]_+, \quad q_{ij}(\alpha) := \sum_{k=1}^d \alpha_k (\mu_k^{(i)} - \mu_k^{(j)})^2, \quad s_{ij}(\alpha) := \max\{\alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)}\},$$

on

$$A := \{\alpha \in [0, 1]^d : \alpha^\top \mathbf{1} = m\},$$

and assume

$$\underline{s} := \min_{\alpha \in A} \min_{(i,j) \in L} s_{ij}(\alpha) > 0, \quad \bar{s} := \max_{\alpha \in A} \max_{(i,j) \in L} s_{ij}(\alpha) < \infty.$$

Further, let

$$\eta^* := \min_{\alpha \in A} \max_{(i,j) \in L} [c s_{ij}(\alpha) - q_{ij}(\alpha)]_+,$$

and let α_η be an optimizer. Then, with

$$F^* := \min_{\alpha \in A} F(\alpha),$$

the point α_η satisfies

$$F(\alpha_\eta) \leq \frac{|L| \eta^*}{\underline{s}}, \quad F^* \geq \frac{\eta^*}{\bar{s}}.$$

Consequently,

$$F(\alpha_\eta) \leq |L| \frac{\bar{s}}{\underline{s}} F^*.$$

If, in addition, F is G -Lipschitz on A with respect to $\|\cdot\|_1$, then for every minimizer $\alpha^* \in \arg \min_A F$,

$$F(\alpha_\eta) \leq F^* + G \|\alpha_\eta - \alpha^*\|_1.$$

Hence, if $F^* \geq \delta > 0$ and $\|\alpha_\eta - \alpha^*\|_1 \leq \varepsilon \delta / G$, then

$$F(\alpha_\eta) \leq (1 + \varepsilon) F^*.$$

Proof. For every $(i, j) \in L$,

$$[c s_{ij}(\alpha_\eta) - q_{ij}(\alpha_\eta)]_+ \leq \eta^*,$$

so

$$F(\alpha_\eta) = \sum_{(i,j) \in L} \frac{[c s_{ij}(\alpha_\eta) - q_{ij}(\alpha_\eta)]_+}{s_{ij}(\alpha_\eta)} \leq \sum_{(i,j) \in L} \frac{\eta^*}{\underline{s}} = \frac{|L| \eta^*}{\underline{s}}.$$

Also, for any $\alpha \in A$,

$$F(\alpha) = \sum_{(i,j) \in L} \frac{[c s_{ij}(\alpha) - q_{ij}(\alpha)]_+}{s_{ij}(\alpha)} \geq \frac{\max_{(i,j) \in L} [c s_{ij}(\alpha) - q_{ij}(\alpha)]_+}{\bar{s}}.$$

Taking the minimum over $\alpha \in A$ gives

$$F^* \geq \frac{\eta^*}{\bar{s}}.$$

Combining the two bounds yields

$$F(\alpha_\eta) \leq |L| \frac{\bar{s}}{\underline{s}} F^*.$$

If F is G -Lipschitz, then

$$F(\alpha_\eta) \leq F(\alpha^*) + G \|\alpha_\eta - \alpha^*\|_1 = F^* + G \|\alpha_\eta - \alpha^*\|_1.$$

The final claim follows from $F^* \geq \delta$ and $\|\alpha_\eta - \alpha^*\|_1 \leq \varepsilon \delta / G$. $\square$

H Rewrite of Approx

I Approximation in fixed dimension

Consider the relaxed problem

$$\begin{aligned}
& \min_{\alpha, \beta} \quad \sum_{(i,j) \in L} \beta_{ij} \\
& \text{s.t.} \quad \sum_{k=1}^d \alpha_k (\mu_k^{(i)} - \mu_k^{(j)})^2 \geq (c - \beta_{ij}) \max\{\alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)}\}, \quad \forall (i,j) \in L, \\
& \quad \beta \geq 0, \quad \alpha \in [0, 1]^d, \quad \alpha^\top \mathbf{1} = m.
\end{aligned} \tag{17}$$

Let

$$\mathcal{A} := \{\alpha \in [0, 1]^d : \alpha^\top \mathbf{1} = m\}.$$

For $(i, j) \in L$, define

$$q_{ij}(\alpha) := \sum_{k=1}^d \alpha_k (\mu_k^{(i)} - \mu_k^{(j)})^2, \quad s_{ij}(\alpha) := \max\{\alpha^\top \sigma^{(i)}, \alpha^\top \sigma^{(j)}\}.$$

Eliminating the slacks yields

$$F(\alpha) := \sum_{(i,j) \in L} \left[c - \frac{q_{ij}(\alpha)}{s_{ij}(\alpha)} \right]_+, \quad F^* := \min_{\alpha \in \mathcal{A}} F(\alpha). \tag{18}$$